INCH-POUND
MIL-PRF-3098/87H
w/AMENDMENT 1
16 August 2023
SUPERSEDING
MIL-PRF-3098/87H
1 November 2017

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR111/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 17 MHz to 62 MHz; third mechanical overtone; noncontrolled; series resonance.

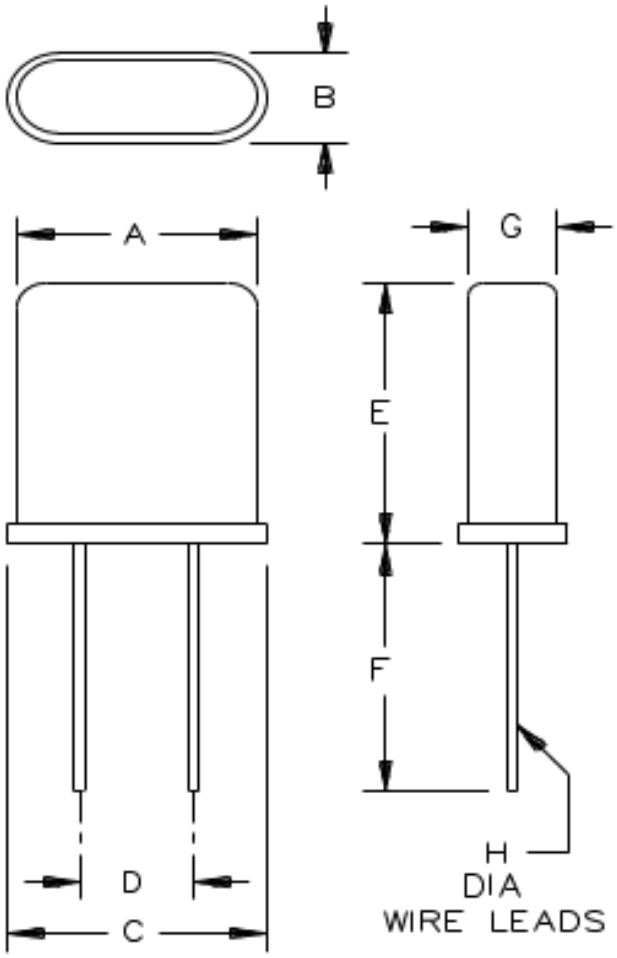


FIGURE 1. Crystal unit - CR111/U.

AMSC N/A
FSC 5955
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Ltr	inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR111/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 17 MHz to 62 MHz, inclusive.

Capacitance, shunt: 4 pF, maximum.

Mode of oscillation: Third mechanical overtone.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Reference temperature: +30°C ±1°C.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Frequency tolerances:

Operating temperature range: ±50 ppm.

Reference temperature: ±15 ppm.

Equivalent resistance: 40 ohms, maximum.

Shock (specified pulse): MIL-STD-202-213, with the following exceptions: 18 impact shocks of 5 gravity units each, time duration 11 milliseconds ±1 millisecond. Three shocks shall be applied in each direction along each of three mutually perpendicular planes.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Vibration: MIL-STD-202-204, test condition A; 0.03 inch double amplitude or 5 gravity units, whichever is less.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent or 2 ohms, whichever is greater.

Aging:

Frequency change permitted: ±5 ppm.

Qualification: Not required. First article testing (FAT) shall be performed according to the qualification table II of MIL-PRF-3098; for product level B, Table II is applicable and for product level S, Table II-A is applicable. Thirty samples for product level B and forty-five samples for product level S shall be selected. Approval of FAT shall be at the discretion of the acquiring activity.

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Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204 MIL-STD-202-213

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:

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(Project 5955-2023-002)

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